

1

2

3

4

5

6

A

B

C

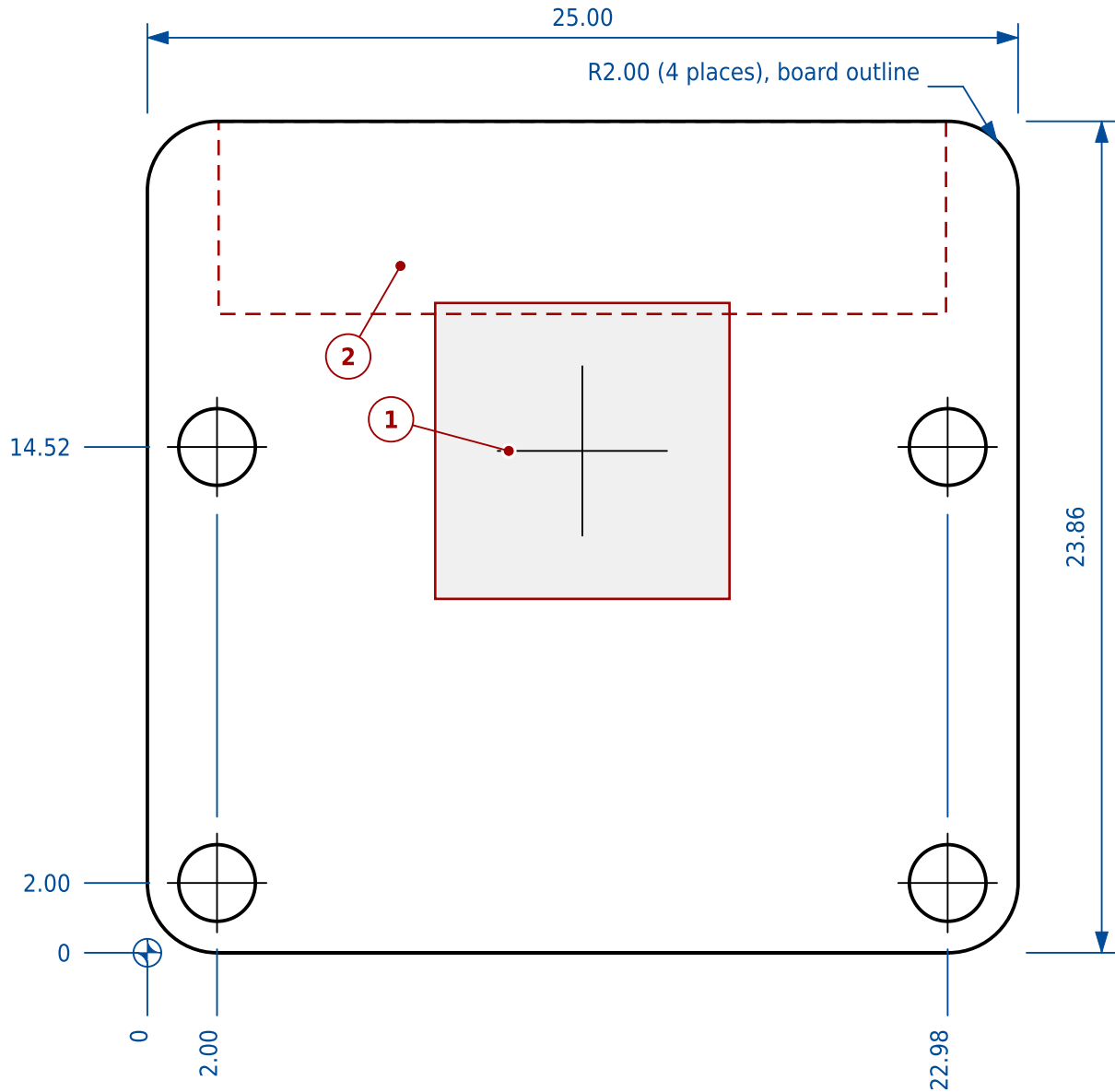
D

A

B

C

D



HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	2.00	2.00	2.20	not given
MT2	22.98	2.00	2.20	not given
MT3	2.00	14.52	2.20	not given
MT4	22.98	14.52	2.20	not given

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	Lens and sensor module	8.26 to 16.71	10.16 to 18.65
2	Camera FFC connector, 15-way	2.05 to 22.93	18.33 to 23.85

LEGEND

- Board outline
- Component body, scheduled
- On the underside, seen through the board
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Hole IDs are this drawing's, the same hole on every camera sheet. The two drawings are read separately and their hole patterns agree to 0.025, which is why one sheet prints 22.98 where the other prints 23.00.
- Optical axis at X 12.49, Y 14.40, the centre of feature 1. Feature 1 is dimensioned 8.5 square. Its lower body, stepped, reaches to 2.76 from the lower edge across 6.72 to 16.14.
- The source drawing is a plan view only and gives no height.
- Source read as a 1.5055:1 plot: scale from the 21.0 x 12.5 hole rectangle, overall dimensions then back within 0.031.
- Every figure printed on the source is an outlined path, not text: its geometry is machine-read, its printed dimensions are transcribed by eye.
- The component side is the lens side; the connector is on the far face.

SOURCES

- Mechanical drawing: <https://datasheets.raspberrypi.com/camera/camera-module-2-mechanical-drawing.pdf> - Raspberry Pi Ltd, RPI-CAM-V2_1, 12/11/2015
- Mounting holes: <https://datasheets.raspberrypi.com/camera/camera-module-2-mechanical-drawing.pdf> - Drawing note, quoted: 4x 2.2mm diameter holes. No land or keep-out is shown.

RASPBERRY PI CAMERA MODULE 2

25 x 23.862 mm, Sony IMX219

DRAWN generated		VERSION v0.0-400-g80d77dd		UNITS mm		SCALE 5:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO RPICAM-2				REV A
MATERIAL PCB, thickness not stated			GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08				

1

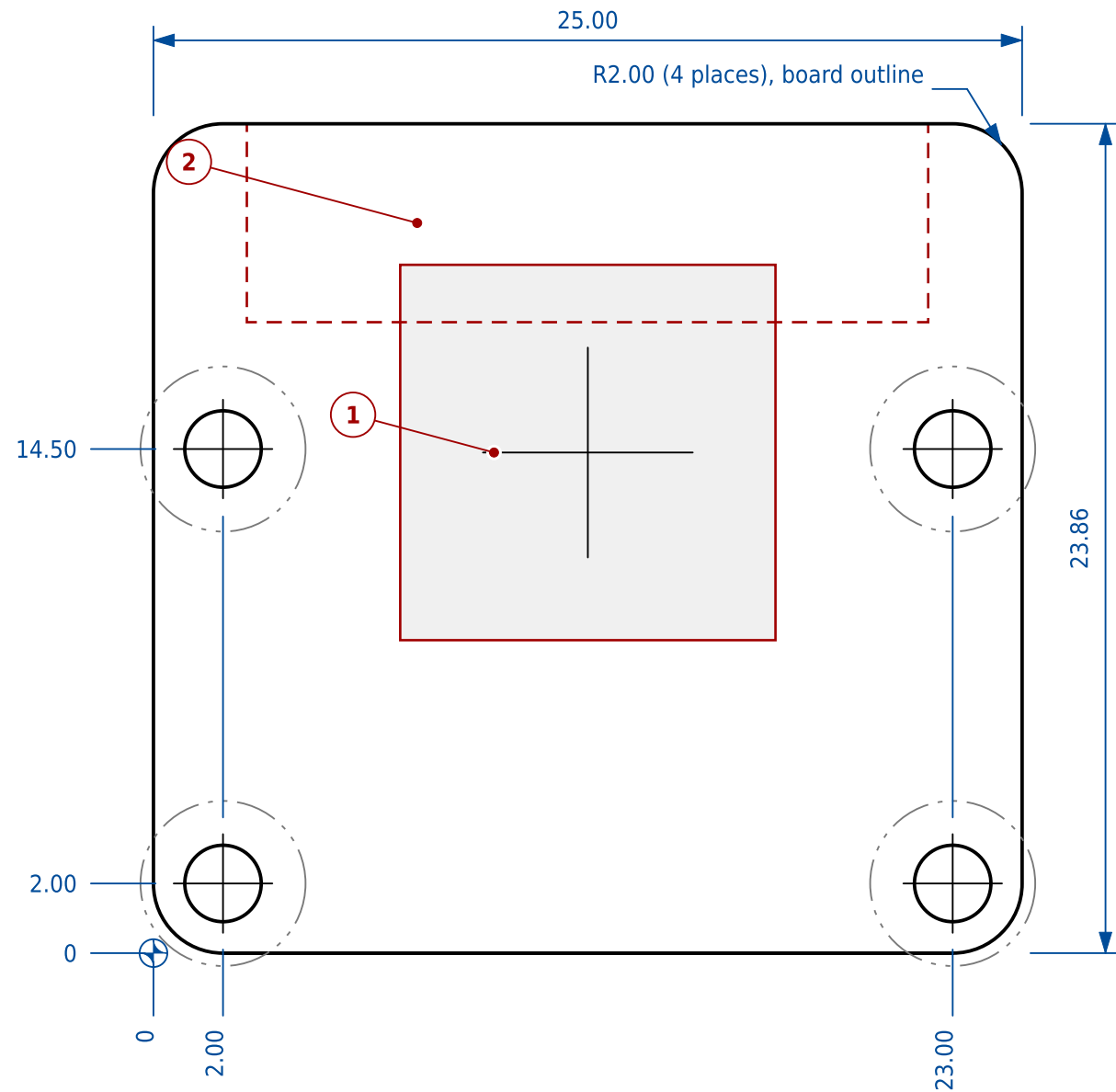
2

3

4

5

6



HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	2.00	2.00	2.20	4.75
MT2	23.00	2.00	2.20	4.75
MT3	2.00	14.50	2.20	4.75
MT4	23.00	14.50	2.20	4.75

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	Lens and sensor module	7.10 to 17.90	9.00 to 19.80
2	Camera FFC connector, 15-way	2.69 to 22.30	18.15 to 23.86

LEGEND

- Board outline
- Component body, scheduled
- On the underside, seen through the board
- Hole keep-out
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Hole IDs are this drawing's, the same hole on every camera sheet. The two drawings are read separately and their hole patterns agree to 0.025, which is why one sheet prints 22.98 where the other prints 23.00.
- Optical axis at X 12.50, Y 14.40, the centre of feature 1. Clear aperture printed o5.75 standard, o6.95 wide; the wide drawing draws o7.00. Feature 1's lower body, 8.9 across, reaches to 1.70 from the lower edge.
- Overall thickness printed 11.3 standard, 12 wide, lens tip to connector back; lens assembly 6.98 and 8.3 above the board. The source's own elevations scale 1.2 short, so no height is drawn here.
- Source read as a true 1:1 plot: scale from the 21.0 x 12.5 hole rectangle, overall dimensions then back within 0.004.
- The standard and wide drawings agree on the outline, holes and lens module within 0.05, so one sheet covers both.

NOTES (continued)

- The component side is the lens side; the connector is on the far face.
- MATERIAL gives the thickness the source states, 1.12 mm, not rounded to a standard finished thickness.

SOURCES

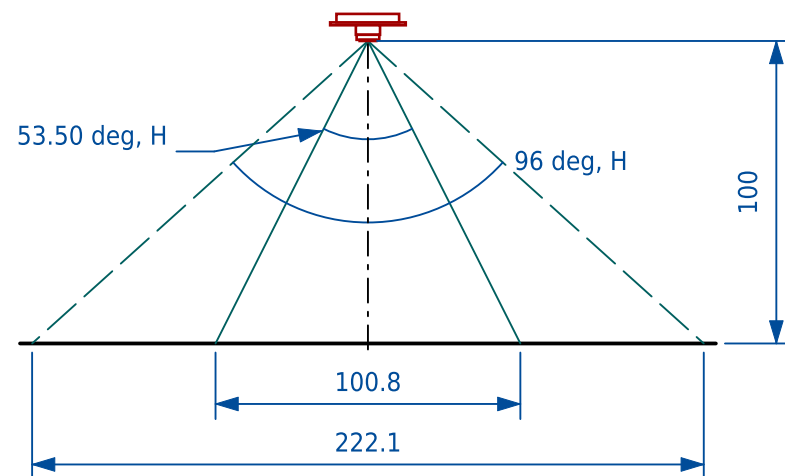
- Mechanical drawing: <https://datasheets.raspberrypi.com/camera/camera-module-3-standard-mechanical-drawing.pdf> - Raspberry Pi Ltd, RP-008153-DS-1
- Mechanical drawing: <https://datasheets.raspberrypi.com/camera/camera-module-3-wide-mechanical-drawing.pdf> - Raspberry Pi Ltd, RP-008155-DS-1
- Mounting holes: <https://datasheets.raspberrypi.com/camera/camera-module-3-standard-mechanical-drawing.pdf> - o2.2, with an o4.75 land round each one.

RASPBERRY PI CAMERA MODULE 3

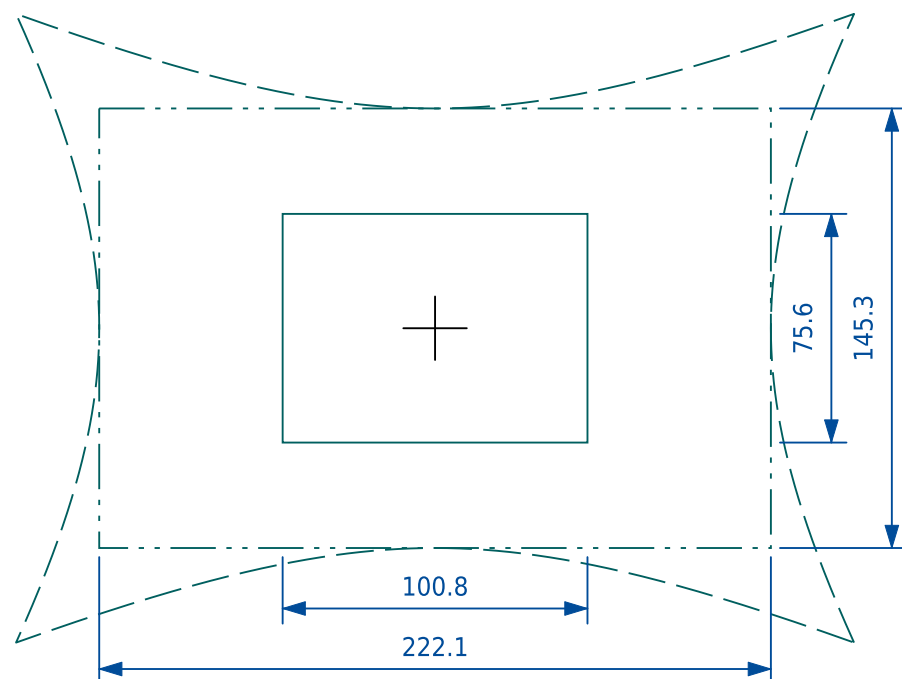
25 x 23.862 mm, standard and wide, Sony IMX708

DRAWN generated		VERSION v0.0-400-g80d77dd	UNITS mm	SCALE 5:1	
SIZE A3	SHEET 1 OF 1	DRAWING NO RPICAM-3			REV A
MATERIAL PCB, 1.12 as given		GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08			

ELEVATION ACROSS THE LONG SIDE



PLAN: THE PICTURE ON A BOARD 100 BELOW



NOTES

1. 65: rectilinear; its 53.50 x 41.41 come back out of 3.60 mm on the 3.6288 x 2.7216 mm active array to 0.01 deg. The 65 it is sold as is the diagonal, 64.42 on the array and 64.94 on OmniVision's image area.
2. AF: the AliExpress "AF-65 Degrees" module used here, a v1.3 clone board, not an Arducam. Its one figure is the 65, the stock lens's diagonal; its angles, f and F are the stock lens's, ASSUMED.
3. 120: the page's 120 is the DIAGONAL, so the catalogue's 120 x 90 is REJECTED: no lens sees as far across as to the corner. Its 96 x 72, for the camera without IR filter, is the diagonal x 0.8 and 0.6 like its neighbours: an equidistant split, not a measurement. YXF print D as H, H as V, V as D. The margin absorbs every row marked margin.

NOTES (continued)

4. DISTORTION: a ray A/2 off the axis meets a board Z below at Z tan(A/2) whatever the lens did to it. The dashed outline is the sensor's edge walked onto the board equidistantly: its sides bow outwards, so the rectangle the angles give, which every height is worked from, is inside the picture: its corner reaches 138.6, 103.9 against the rectangle's 111.1, 72.7. At the long side's edge a pixel covers 2.23 x the board it does on the axis.
5. DEPTH OF FIELD, DERIVED, thin lens, circle of confusion 2 px, 2.8 um: hyperfocal $f^2 / (N c) + f$, sharp from half of it to infinity focused there. Raspberry Pi's "Approx 1 m to infinity" is a 1.6 px circle.

SOURCES

1. Raspberry Pi camera documentation; OmniVision OV5647 datasheet; Arducam 5MP OV5647 documentation; Arducam B006604 product page; Arducam B0121 product page, the B0176's predecessor; Arducam B0176 on UCTRONICS, Arducam's own store; Arducam motorized focus camera; Commonlands, OV5647 lens table; YXF YXF4Y001A1 M6 lens for OV5647 modules; Waveshare RPi Camera (G); AliExpress listing of the autofocus module used here; Raspberry Pi forums, "setting lens focus", June 2013; Raspberry Pi forums, "Camera with Macro focus", November 2013; Arducam B0031, OV5647 with an M12 lens focused by hand. Addresses, pinned Internet Archive captures where one exists, in raspberry_pi_camera/optics.py; every quote checked against the cached page by raspberry_pi_camera/verify_optics.py, in ASCII here: x, um, deg and infinity for the pages' signs.

FIELD OF VIEW, deg, EVERY FIGURE

	FIGURE		H	V	DIAG	
65	Raspberry Pi	DECL	53.50	41.41	--	USED
	Arducam B0033	DECL	54	41	--	margin
	3.60 mm on the active array	DERI	53.50	41.41	64.42	agrees
AF	AliExpress, AF-65	DECL	--	--	65	
	the stock lens's pair	ASSU	53.50	41.41	--	USED
120	Arducam B006604 page	DECL	--	--	120	
	Arducam catalogue, B006604	DECL	120	90	--	REJECTED
	Arducam catalogue, B006604N	DECL	96	72	--	USED
	120 diagonal, equidistant	DERI	96	72	120	agrees
	120 diagonal, rectilinear	DERI	108.36	92.20	120	margin
	120 diagonal, equisolid	DERI	94.31	69.83	120	margin
	Commonlands CIL282, 2.2 mm	DECL	96	72	122	agrees
	Commonlands CIL282, scaled to 120	DERI	94.43	70.82	120	margin
	YXF4Y001A1, 1.79 mm	DECL, relabelled	92.40	73.90	119.90	margin

FOCUS; HYPERFOCAL AND NEAR AT 2 / 1 px

	f mm	F	FOCUS	RANGE, mm	HYPER m	NEAR m
65	3.60 DECL	F2.9 DECL	Fixed	1000 to inf	1.60 / 3.20	0.80 / 1.60
AF	3.60 ASSU	F2.9 ASSU	Motorized	not published	1.60 / 3.20	0.80 / 1.60
120	2.17 DERI	F2.4 ASSU	Fixed	1000 to inf	0.70 / 1.40	0.35 / 0.70
WS G	3.15 DECL	F2.35 DECL	Adjustable	100 to inf	--	--
v1.3 unscr.	--	--	By hand	60 to inf forum	--	--
B0031	--	--	By hand	25.4 to inf DECL	--	--
B0176	--	--	Motorized	80 to inf DECL	--	--

AF, IF FOCUSED AT Z: SHARP FROM, TO, mm

Z	NEAR	FAR	DEPTH
30	29.5	30.5	1.0
80	76.3	84.0	7.7
150	137.4	165.1	27.8

LEGEND

- The 65 deg lens: rays, and its picture on the board
- - - The 120 deg lens: rays, and its picture on the board
- - - The rectangle the 120 deg lens's angles give
- Camera Module v1.3
- Dimension, extension and leader

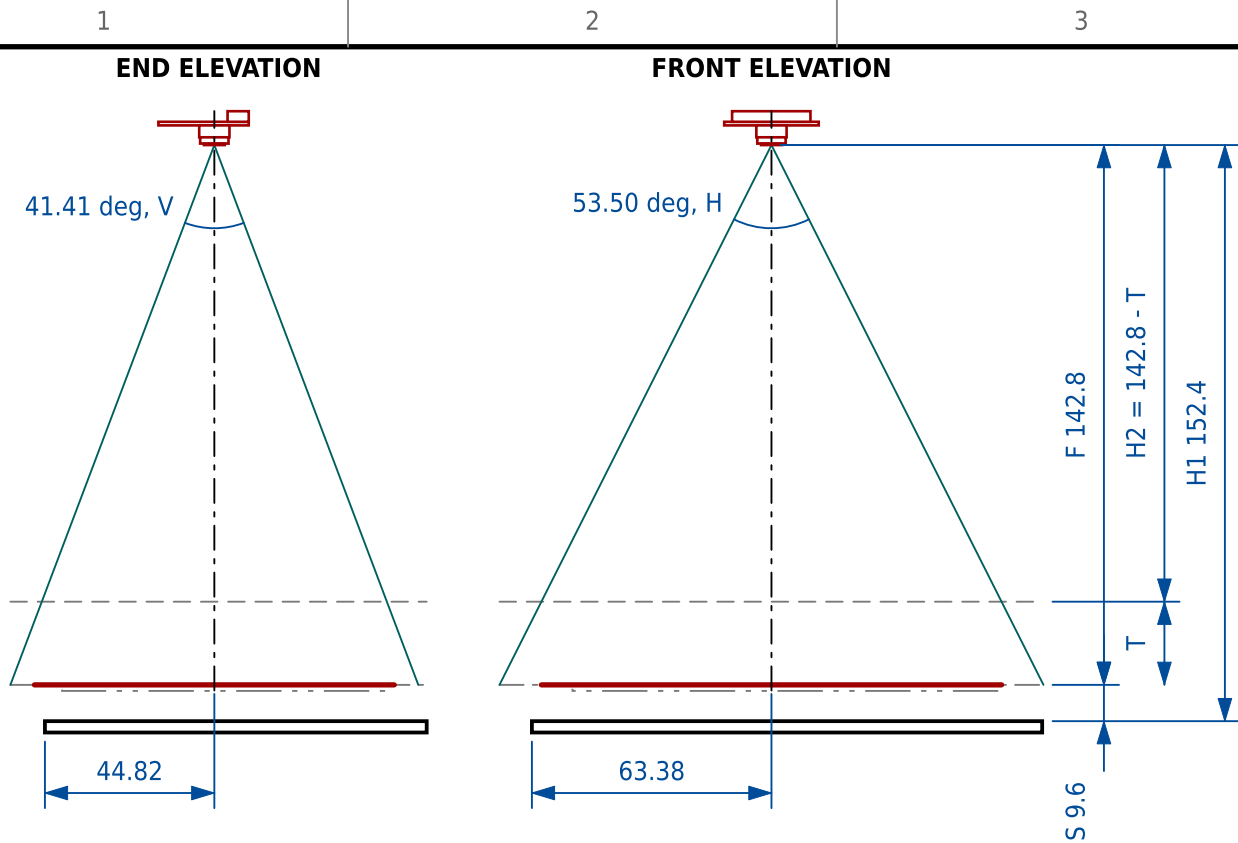
OV5647 LENSES AND FOCUS

Camera Module v1: stock, autofocus and 120 degree lenses

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-485-g9e53917	mm	1:2.5
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	RPICAM-LENS	A
SUBJECT	GENERAL TOLERANCE	PROJECTION	
OV5647 lenses	angles as declared, DERI and ASSU as flagged		

NOTES (continued)

6. FIXED FOCUS, ASSUMED at 2 m, the stock lens's hyperfocal distance; the 120's is 0.70 m, so its 1 m reads copied. At 100 mm a point spreads to 30 px (1.1 mm on the board) at 65 and 13 px (0.8 mm) at 120: out of focus, and within two pixels the same set at infinity.
7. AF: no close limit is published. The others, each its own module's: the v1.3's lens unscrewed by hand "about 6 cm", 3 cm as it falls out, by forum users, Raspberry Pi publish none; the B0031 "less than an inch"; the B0176, a different motorised module, "80mm to infinity".
8. WS G, Waveshare's RPi Camera (G): 160 deg diagonal on Waveshare's page, 120 across on The Pi Hut's; its 3.15 mm gives 82.5 diagonal equidistantly, so it does not hold together and is not drawn.



- NOTES**
1. First angle; the end elevation is seen from the right. One camera: a Raspberry Pi Camera Module v1.3, stock 65 deg lens, looking straight down, reading the whole sensor.
 2. Heights are to the lens FACE. The entrance pupil the optics are worked to is ASSUMED behind it by at most the lens's 5.20 mm, so the picture is up to 5.2% larger, never smaller, and the boards no nearer than the close limit.

X0 Y0

- NOTES (continued)**
3. F is, per frame, the higher of two heights: the one that just frames it with the lens focused there, frame A $139.2 + f\ 3.60 = 142.8$, frame B $100.1 + f\ 3.60 = 103.7$ (DERIVED, thin lens), and the lens's close limit unscrewed, 60 mm: "About the closest you can get this way is about 6 cm." (a Raspberry Pi forum user, 2013; REPORTED). The framing governs every frame.
 4. $H1 = S + F$, frame A 152.4, frame B 113.3, where S is the mounting plate's face, the standoffs' base, to the demo board's top face: 9.6 ± 0.04 , M3 x 8 mm standoffs (TT-MP-PLATE) and a 1.56 to 1.60 mm board (each revision's KiCad file): 9.56 to 9.60, F set from the higher. Neither states a fabrication tolerance.
 5. $H2 = F - T$, the clearance, frame A $142.8 - T$, frame B $103.7 - T$, where T is the demo board's top face to the tallest part on it, on whichever revision is fitted: MEASURE, not published.
 6. HEADROOM: a part standing on a board, anywhere in its envelope stays in frame A's picture, the lens refocused to F, up to 13.6 mm above the board's top face (DERIVED). If T is more, raise the camera.

- NOTES (continued)**
7. REFOCUS THE LENS TO F, 142.8 mm from the lens face to the boards for frame A, 103.7 for frame B. As sold it is set far -- "Approx 1 m to infinity" (Raspberry Pi; infinity is their sign), "from about 0.5m to infinity" (raspi.tv) -- and at frame A's F a point spreads to 21 px (DERIVED). raspi.tv adds a +2D lens in front, "focus at about 25cm", too far for F; a reader's comment there: "You CAN change the focus of the stock lens on the pi camera. It is tricky but can be done." Unscrew it until the boards are sharp.
 8. A frame is the sensor's own 4:3 round its target plus 5.00 mm all round, turned whichever way needs the lower camera; the table gives which, and where the lens axis goes over each.
 9. Frame A, Every board, any revision: Every revision's assembled envelope -- outline, connectors, Pmod bodies -- in plate coordinates, in one: a camera fixed over the plate serves them all.
 10. Frame B is the union of every LED and 7-segment over all five revision families. They are not clustered -- 90.91×62.79 of a 135×101 plate -- so frame B buys little over frame A.

LENS FACE HEIGHTS, mm: BOARDS IN VIEW AND IN FOCUS

	THE LENS FACE ABOVE	FRAME A	FRAME B	TERM
H1	the mounting plate's face, the standoffs' base	152.4	113.3	S 9.6 ± 0.04
H2	the highest point on the board, as clearance	$142.8 - T$	$103.7 - T$	T: MEASURE
F	the demo board's top face, any revision	142.8	103.7	DERIVED

FRAMES, AND WHERE THE LENS AXIS GOES, mm

	FRAME	RECTANGLE	LONG	AXIS X	AXIS Y
A	Every board, any revision	140.27×105.20	X	63.38	44.82
B	Every LED and 7-seg	100.91×75.69	X	67.13	53.12

LEGEND

- Subject outline
- LED, display or connector
- - - Adjacent part or connector body
- Frame A's target, on the plane F is measured from
- The 65 deg lens's field of view, 53.50×41.41 , to its picture's edge
- - - Frame A, the rectangle the picture must cover
- - - Frame B, the rectangle the picture must cover
- The mounting plate's face, S below the board's top face
- - - The highest point on the board, T above the board's top face: NOT TO SCALE, to measure
- Dimension, extension and leader

NOTES (continued)

11. T: no board file gives a part a height, and nobody publishes one for every revision. Measure it on the boards in use, from the board's top face to the top of its tallest part.

SOURCES

1. Plate geometry: tinytapeout/mounting_plate/plate.py - Outline, placements and standoff; see TT-MP-PLATE.
2. Board and indicator positions: tinytapeout/boards.py - Outlines, connectors, Pmod bodies, LEDs and 7-segment displays, in plate coordinates, and each board's thickness.
3. raspi.tv, 25 May 2013: <https://web.archive.org/web/20260212013652/https://raspi.tv/2013/adapt-your-raspberry-pi-camera-for-close-up-use>
4. Lens figures and focus: RPICAM-LENS - Every lens figure on this sheet, declared and derived, with the vendor pages it is quoted from.

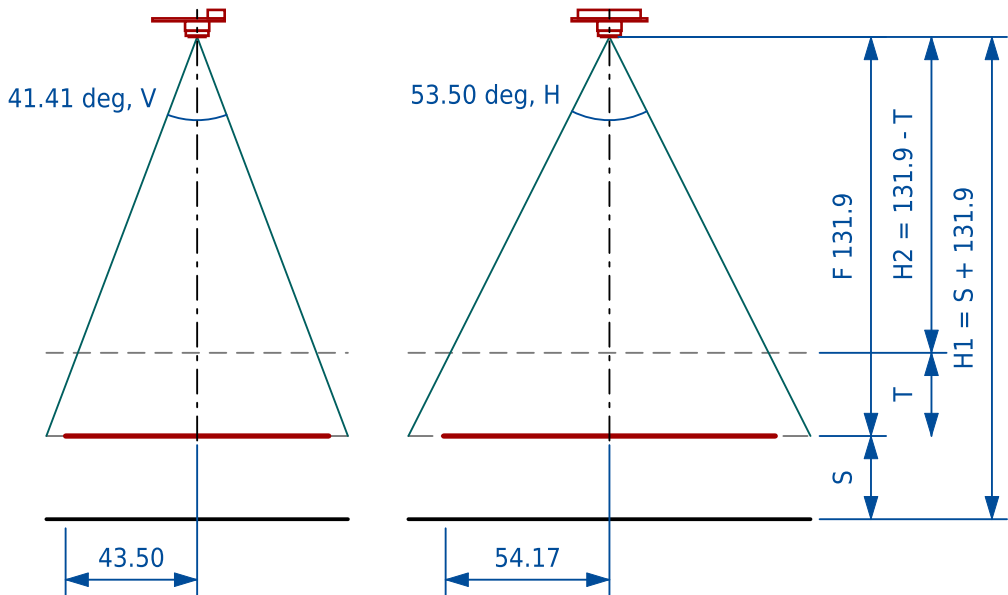
CAMERA OVER THE TT MOUNTING PLATE

Camera Module v1.3, stock 65 degree lens: lens face heights

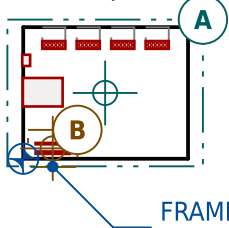
DRAWN generated		VERSION v0.0-536-g6b746d3	UNITS mm	SCALE 1:2, PLAN 1:2.5
SIZE A3	SHEET 1 OF 1	DRAWING NO RPICAM-OVER-PLATE		REV A
SUBJECT TT Mounting Plate		GENERAL TOLERANCE plate ± 0.20, LEDs ± 0.10, heights DERIVED		PROJECTION

END ELEVATION

FRONT ELEVATION



PLAN, 1:5



FRAMES A AND B: edges 0.43 apart, drawn as one line

NOTES

1. First angle; the end elevation is seen from the right. One camera: a Raspberry Pi Camera Module v1.3, stock 65 deg lens, looking straight down, reading the whole sensor.

NOTES (continued)

2. Heights are to the lens FACE. The entrance pupil the optics are worked to is ASSUMED behind it by at most the lens's 5.20 mm, so the picture is up to 9.2% larger, never smaller, and the board and its LEDs no nearer than the close limit.
3. F is, per frame, the higher of two heights: the one that just frames it with the lens focused there, frame A $128.3 + f\ 3.60 = 131.9$, frame B $32.4 + f\ 3.60 = 36.0$ (DERIVED, thin lens), and the lens's close limit unscrewed, 60 mm: "About the closest you can get this way is about 6 cm." (a Raspberry Pi forum user, 2013; REPORTED). The forum does not say whether its "about 6 cm" is from the lens face; it is taken as from the face, note 2's convention.
4. $H1 = S + F$, frame A $S + 131.9$, frame B $S + 60.0$, where S is what the Arty stands on, under its rubber feet, to its top face: a foot and the board: MEASURE, not published.
5. $H2 = F - T$, the clearance, frame A $131.9 - T$, frame B $60.0 - T$, where T is the Arty's top face to the top of its tallest part: MEASURE, not published.
6. HEADROOM: a part standing on the board, anywhere in its envelope stays in frame A's picture, the lens refocused to F, up to 13.6 mm above the board's top face at the envelope's corner, the worst case; where J9 stands, up to 20.9 (DERIVED). If T's part stands higher than its figure, raise the camera.

NOTES (continued)

7. REFOCUS THE LENS TO F, 131.9 mm from the lens face to the board and its LEDs for frame A, 60.0 for frame B. As sold it is set far -- "Approx 1 m to infinity" (Raspberry Pi; infinity is their sign), "from about 0.5m to infinity" (raspi.tv) -- and at frame A's F a point spreads to 23 px (DERIVED). raspi.tv adds a +2D lens in front, "focus at about 25cm", too far for F; a reader's comment there: "You CAN change the focus of the stock lens on the pi camera. It is tricky but can be done." Unscrew it until the board and its LEDs are sharp.
8. A frame is the sensor's own 4:3 round its target plus 5.00 mm all round, turned whichever way needs the lower camera; the table gives which, and where the lens axis goes over each.
9. Frame A, Whole board: The assembled envelope: the outline together with the connectors and Pmod bodies that overhang it.
10. Frame B, LD0-LD7: The four tri-colour LEDs LD0-LD3 and the four single LD4-LD7, in two rows on a 7.00 mm pitch.
11. Which LED row is which is not named by any Digilent source; FPGA-ARTY-A7 takes the row nearest the edge as the tri-colour LD0-LD3 by package size. Either way both rows are inside the frame, so the framing does not turn on it.
12. S: the A7 drawing has four corner circles, 10.0 mm across scaled from its plot, taken as the rubber feet, and no height for them or the board. Digilent's 3D model of the Arty Rev C, not the A7, has a 1.50 board on feet 3.70 tall, 5.20 in all, but its feet are 8.70 across and it states no tolerance. Measure S on the A7, from what it stands on to its top face.

LENS FACE HEIGHTS, mm: BOARD IN VIEW AND IN FOCUS

	THE LENS FACE ABOVE	FRAME A	FRAME B	TERM
H1	what the Arty stands on, under its feet	$S + 131.9$	$S + 60.0$	S: MEASURE
H2	the highest point on the board, as clearance	$131.9 - T$	$60.0 - T$	T: MEASURE
F	the Arty's top face	131.9	60.0	DERIVED

FRAMES, AND WHERE THE LENS AXIS GOES, mm

	FRAME	RECTANGLE	LONG	AXIS X	AXIS Y
A	Whole board	129.33 x 97.00	X	54.17	43.50
B	LD0-LD7	32.68 x 24.51	X	19.50	6.83

LEGEND

- Subject outline
- LED, display or connector
- Adjacent part or connector body
- Frame A's target, on the plane F is measured from
- The 65 deg lens's field of view, 53.50 x 41.41, to its picture's edge
- Frame A, the rectangle the picture must cover
- Frame B, the rectangle the picture must cover
- What the Arty stands on, S below the board's top face: NOT TO SCALE, to measure
- The highest point on the board, T above the board's top face: NOT TO SCALE, to measure
- Dimension, extension and leader

NOTES (continued)

13. T: the A7 drawing gives no part a height. The Rev C model's tallest part is the Ethernet jack J9's shield, 14.80 above the board's top face, and the A7 drawing puts J9 in the same place: more than the 13.6 mm headroom at the envelope's corner, less than the 20.9 where J9 stands. Measure T on the A7, from its top face to the top of its tallest part, and which part.

SOURCES

1. Board geometry: fpga/boards.py - Outline, Pmod hosts, connectors and LED rows; see FPGA-ARTY-A7.
2. 3D model, Arty Rev C: https://digilent.com/reference/_media/reference/programmable-logic/artys/artys_rev_c_cad.zip - Digilent, Arty Rev C.stp, 20 June 2016: not the A7. Its heights are quoted in the notes; no height on this sheet is set from it.
3. raspi.tv, 25 May 2013: <https://web.archive.org/web/20260212013652/https://raspi.tv/2013/adapt-your-raspberry-pi-camera-for-close-up-use>
4. Lens figures and focus: RPICAM-LENS - Every lens figure on this sheet, declared and derived, with the vendor pages it is quoted from.

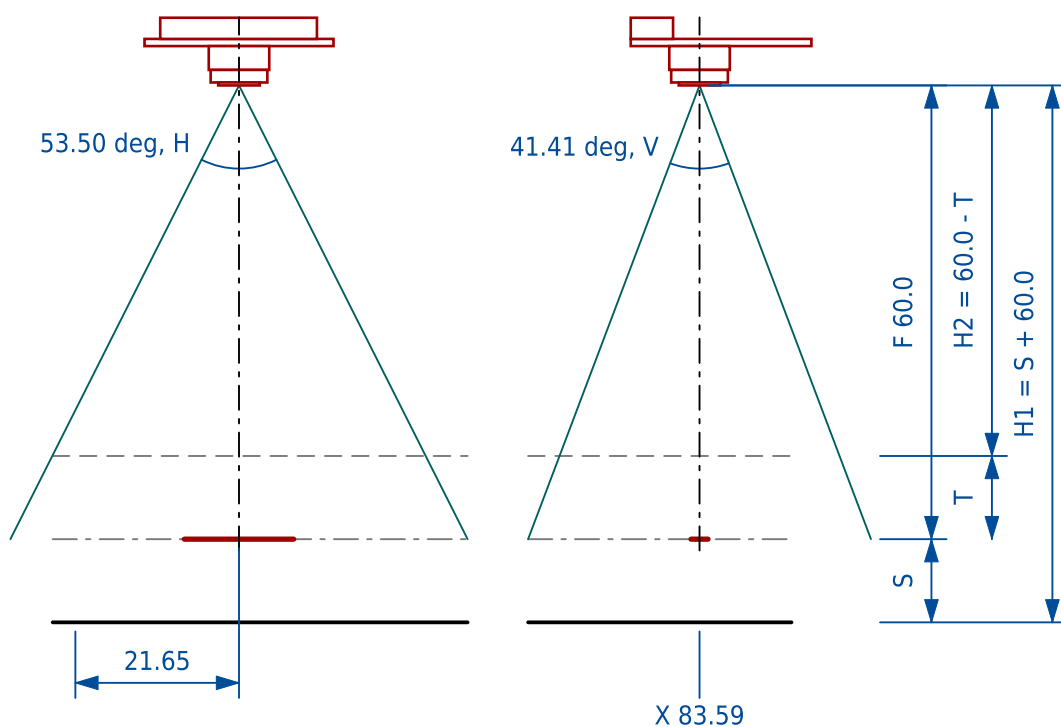
CAMERA OVER THE ARTY A7

Camera Module v1.3, stock 65 degree lens: lens face heights

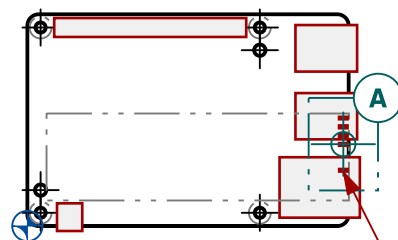
DRAWN generated		VERSION v0.0-606-gd307ee4		UNITS mm		SCALE 1:2.5, PLAN 1:5	
SIZE A3	SHEET 1 OF 1		DRAWING NO RPICAM-OVER-ARTY				REV A
SUBJECT Digilent Arty A7		GENERAL TOLERANCE DXF +/-0.20, plot bodies +/-0.30, heights DERIVED					PROJECTION 

END ELEVATION

FRONT ELEVATION



PLAN, 1:2



NOTES

1. First angle; the end elevation is seen from the right. One camera: a Raspberry Pi Camera Module v1.3, stock 65 deg lens, looking straight down, reading the whole sensor.
2. Heights are to the lens FACE. The entrance pupil the optics are worked to is ASSUMED behind it by at most the lens's 5.20 mm, so the picture is up to 9.2% larger, never smaller, and the LEDs no nearer than the close limit.

LEDs A1-A4 and PWR

NOTES (continued)

3. F 60.0 is the higher of two heights: the one that just frames frame A with the lens focused there, $24.3 + f \ 3.60 = 27.9$ (DERIVED, thin lens), and the lens's close limit unscrewed, 60 mm: "About the closest you can get this way is about 6 cm." (a Raspberry Pi forum user, 2013; REPORTED). The forum does not say whether its "about 6 cm" is from the lens face; it is taken as from the face, note 2's convention.
4. $H1 = S + 60.0$, where S is the mounting plate's face, the base of the standoffs under the Pi 5, to the card's top face: MEASURE, not published.
5. $H2 = 60.0 - T$, the clearance, where T is the card's top face to the assembly's highest point, the heatsink's blower or a part of the HAT: MEASURE, not published.
6. REFOCUS THE LENS TO F, 60.0 mm from the lens face to the LEDs. As sold it is set far -- "Approx 1 m to infinity" (Raspberry Pi; infinity is their sign), "from about 0.5m to infinity" (raspi.tv) -- and at F a point spreads to 52 px (DERIVED). raspi.tv adds a +2D lens in front, "focus at about 25cm", too far for F; a reader's comment there: "You CAN change the focus of the stock lens on the pi camera. It is tricky but can be done." Unscrew it until the LEDs are sharp.

NOTES (continued)

7. Frame A is the sensor's own 4:3 round A1-A4 and PWR plus 5.00 mm all round. A1 to A4 are in a column on the +Y side of the retention screw, PWR on the -Y side; measured off SQRL's own photograph to +/-1.0 mm. At F the picture is larger; the table gives the crop.
8. OBSERVED, reported by Tim Ansell, 3 Oct 2026, of the deployed hardware: the autofocus camera on the reference Acorn, the "AF-65" OV5647 module, is about 10 cm above the Acorn's LEDs, the picture looks in focus, and it needs a digital zoom or crop onto the Acorn's end. DERIVED, the 10 cm approximate, the module's angles ASSUMED the stock lens's: at 100 mm the picture covers 72.9 x 97.2 mm, frame A is a crop of x 3.97, the 14.47 mm LED column 386 px.
9. S is six terms -- standoffs under the Pi, Pi 5 board, Pi-to-HAT spacers, HAT board, M.2 socket seat, card -- and only the card's 0.80 +/-0.08 is published (M.2 specification); the HAT+ specification only recommends "provide at least 15mm board-to-board spacers; 16mm spacers are ideal". Measure S as one figure.

LENS FACE HEIGHTS, mm: THE LEDs IN VIEW AND IN FOCUS

	THE LENS FACE ABOVE	HEIGHT	TERM
H1	the mounting plate's face, the standoffs' base	$S + 60.0$	S: MEASURE
H2	the assembly's highest point, as clearance	$60.0 - T$	T: MEASURE
F	the card's top face, where the LEDs are	60.0	DERIVED

FRAME A AND THE PICTURE AT THE CARD, mm, DERIVED

LENS TO LEDs	WHAT	COVERS X x Y	CROP TO A	LED COLUMN
--	frame A, long side along Y	18.35 x 24.47	x 1.00	--
F 60.0	the v1.3's picture, refocused	42.6 x 56.9	x 2.32	660 px
about 100	the AF-65, as reported; angles ASSUMED	72.9 x 97.2	x 3.97	386 px

LEGEND

- Subject outline
- LED, display or connector
- Adjacent part or connector body, and the Acorn CLE-215+
- Frame A's target, on the plane F is measured from
- The 65 deg lens's field of view, 53.50 x 41.41, to its picture's edge
- Frame A, the rectangle the picture must cover
- The mounting plate's face, S below the card: NOT TO SCALE, to measure
- The assembly's highest point, T above the card: NOT TO SCALE, to measure
- Dimension, extension and leader

NOTES (continued)

10. The camera is over the LEDs, not the card, so that rays to them rise clear of the heatsink's blower beside them: ASSUMED from Enjoy-Digital's photographs, which show the blower there and no height for it (T includes it, unmeasured); confirm when it is measured. The camera board overhangs it, its underside 5.20 above the lens face.

SOURCES

1. Board geometry: raspberry_pi/boards.py - Pi 5 outline and connectors; see RPI-5.
2. Card, LEDs, HAT and stack: accessories/parts.py - The card, its seat, its LEDs, S and T; see ACC-HAT-M2POE.
3. Raspberry Pi HAT+ specification: <https://web.archive.org/web/20260115212335/https://datasheets.raspberrypi.com/hat/hat-plus-specification.pdf>
4. raspi.tv, 25 May 2013: <https://web.archive.org/web/20260212013652/https://raspi.tv/2013/adapt-your-raspberry-pi-camera-for-close-up-use>
5. Lens figures and focus: RPICAM-LENS - Every lens figure on this sheet, declared and derived, with the vendor pages it is quoted from.

CAMERA OVER THE ACORN CLE-215+

Camera Module v1.3, stock 65 degree lens: lens face heights

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-485-g9e53917	mm	1:1, PLAN 1:2
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	RPICAM-OVER-ACORN	A
SUBJECT	GENERAL TOLERANCE	PROJECTION	
Acorn CLE-215+, Pi 5	Pi 5 +/-0.20, LEDs +/-1.0, heights DERIVED		